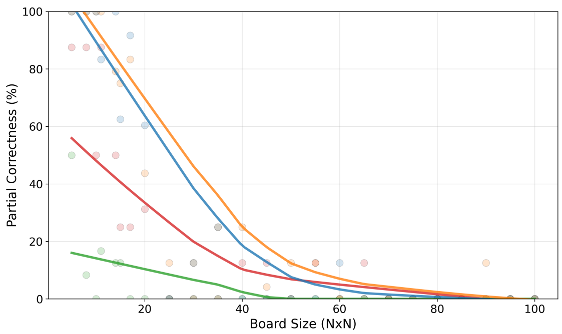
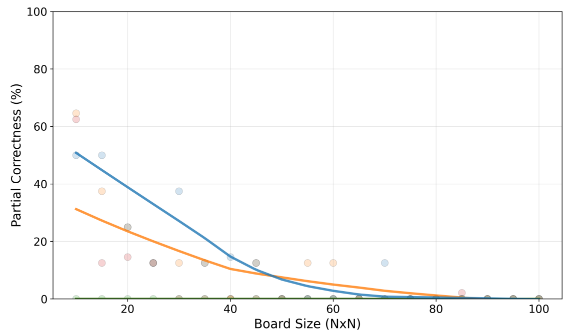
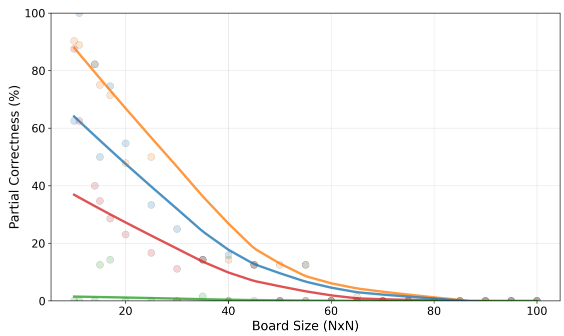
Additionally, models seemed to have trouble identifying words across multiple columns and in unusual orientations. The average accuracy rate was highest for words spelled traditionally, horizontally left-to-right, second-highest backwards spelling, horizontal right-to-left, and lowest for vertical spelling, down-to-up, as shown in Figure 15.



(a) 3 Letter Words

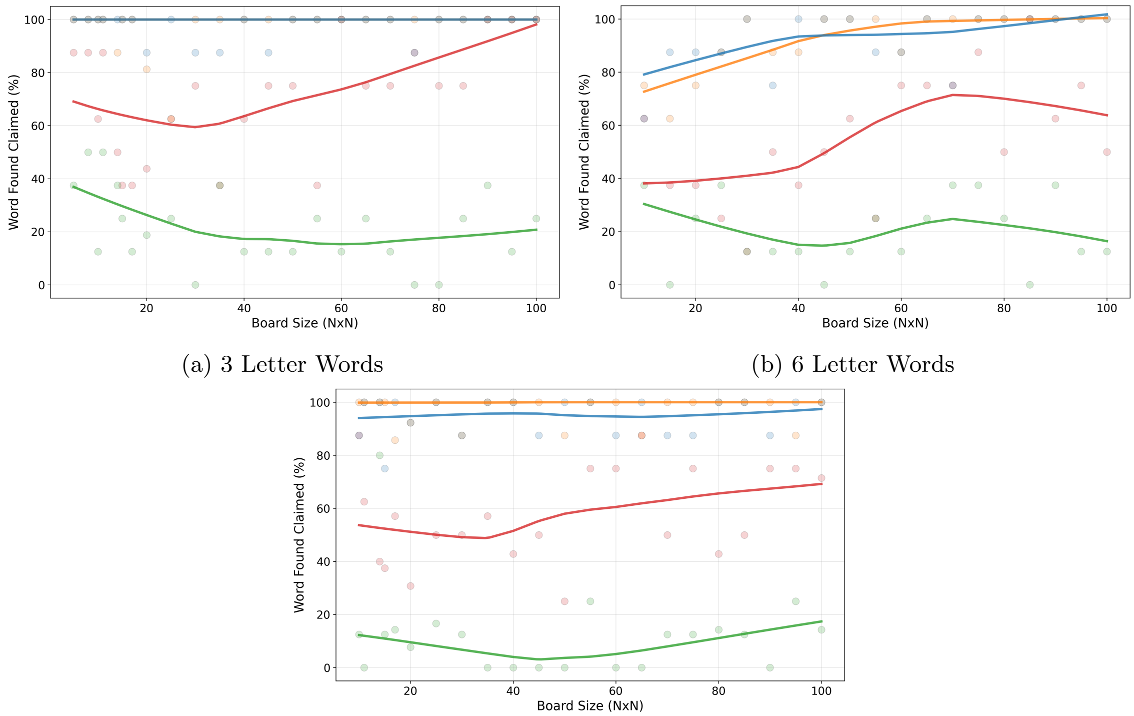


(b) 6 Letter Words



(c) 9 Letter Words

Figure 13: Accuracy vs Grid Size (Search)



(a) 3 Letter Words

(b) 6 Letter Words

(c) 9 Letter Words

Figure 14: Claimed Detection vs Grid Size (Search)

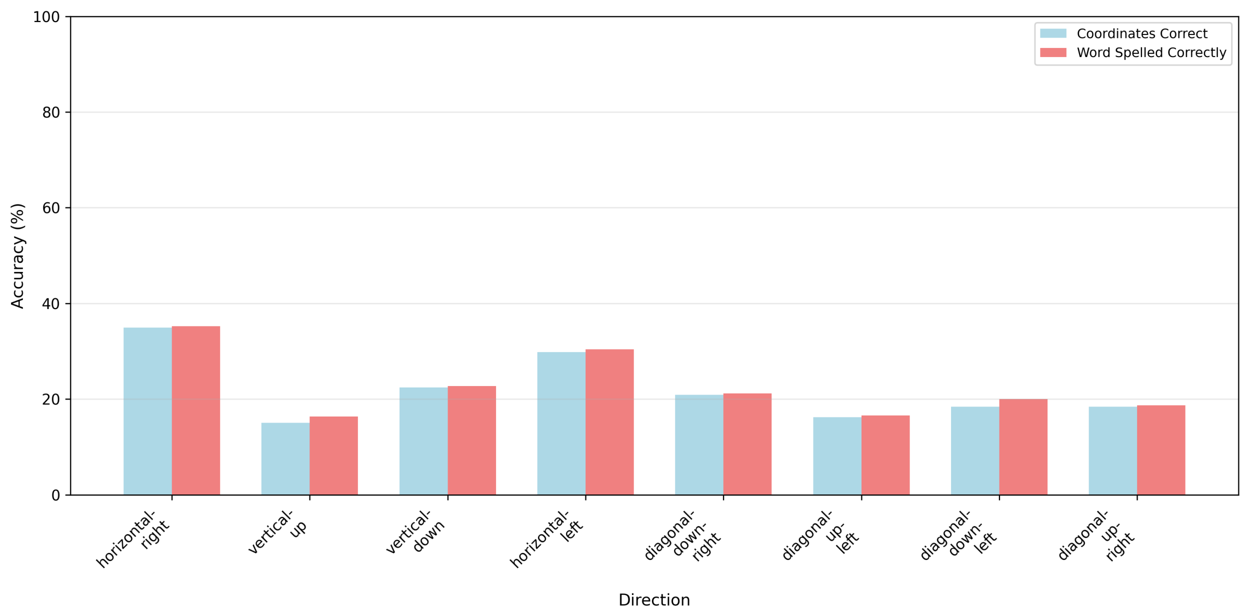


Figure 15: Accuracy by Direction (Search)

4.5 Slide

Models	Smallest 5 Grids	Largest 5 Grids
GPT-4o; 1 slide	28%	8%
GPT-4.1; 1 slide	64%	16%
Claude No Thinking; 1 slide	92%	2%
Claude Medium Thinking; 1 slide	88%	20%
GPT-4o; 2 slides	28%	16%
GPT-4.1; 2 slides	32%	4%
Claude No Thinking; 2 slides	76%	8%
Claude Medium Thinking; 2 slides	88%	24%
GPT-4o; 3 slides	12%	0%
GPT-4.1; 3 slides	44%	0%
Claude No Thinking; 3 slides	64%	12%
Claude Medium Thinking; 3 slides	64%	4%
GPT-4o; 4 slides	0%	8%
GPT-4.1; 4 slides	40%	12%
Claude No Thinking; 4 slides	56%	4%
Claude Medium Thinking; 4 slides	56%	4%
GPT-4o; 5 slides	0%	0%
GPT-4.1; 5 slides	28%	4%
Claude No Thinking; 5 slides	36%	4%
Claude Medium Thinking; 5 slides	44%	0%
GPT-4o; 6 slides	16%	0%
GPT-4.1; 6 slides	28%	0%
Claude No Thinking; 6 slides	32%	0%
Claude Medium Thinking; 6 slides	32%	4%

Table 5: Model Accuracy on Small and Large Grids (Slide)

This final task was designed to test relative geometry and reasoning. Grid sizes were tested from 5×5 to 75×75, with increments of 5×5. Each grid size was tested with 1-6 slides. The final test showed a similar pattern of strong performance on initial, smaller boards and quick performance deterioration as grid sizes grew. The Anthropic models consistently outperformed the OpenAI ones until a grid size of 70×70, after which all accuracy was effectively 0. This deterioration was even quicker than in Word Search, implying that this test was overall more difficult despite the models having a more consistent performance on the smaller boards. When plotting accuracy, each slide of the multi-slide was treated independently to remove cascading effects. The results can be seen in Figure 16.

The number of slides was also a determining factor in the models' average accuracy, with much quicker dropoffs in accuracy for prompts containing more slides. This was to be expected as the introduction of multi-step reasoning on top of the spatial reasoning likely compounded the difficulty. Even when asked to output intermediate steps the models showed no significant increase in performance, suggesting difficulty remembering where they had even placed the X before. A comparison of slides can be seen in Figure 17.

The consistency is somewhat more difficult to parse for this test. The models often ignored walls but it remains unclear whether they miscounted the position of the walls while reading board states or experienced a systemic error with wall detection itself. There were still some classic miscounting issues as well, as shown in Figure 18.